## Digital Signatures & Hashing

1. XYZ University implements an online exam system where students submit assignments and receive digitally signed feedback. The university uses SHA-256 for hashing and RSA for signing.

   a) Explain how digital signatures and hashing help in verifying the integrity and authenticity of feedback.

   b) If the feedback file hash doesn't match the signature, what could be the possible reasons?

2. You are given a message and its corresponding digital signature. How would you verify its authenticity using public key cryptography? What role does a hash function play here?

3. What is the evidentiary value of an email or SMS in a cybercrime case under the Indian Evidence Act? Why are digital signatures important in proving authenticity?

## Cybersecurity Threats & Legal Challenges

4. A company in India was fined for failing to protect user data, which was breached due to inadequate cybersecurity. The breach involved sensitive data and legal issues under the IT Act 2000, RBI Act, IPC, and IPR Act. What steps should the company take to resolve these legal challenges and strengthen its cybersecurity practices?

5. A university's internal network was hit by a ransomware attack, exposing student and research data due to outdated systems. What measures should the university implement to classify and prevent future cyber threats?

6. A financial institution in India suffered a cyberattack through a compromised email, leaking customer data and raising concerns under several legal acts. How should the institution address these issues and enhance cybersecurity awareness?

7. A company employee shared sensitive customer financial details via SMS without consent. Under which sections of the IT Act or IPC might they be charged? Explain with legal references.

## Hashing Techniques

8. A company uses the mid-square method to hash 4-digit employee IDs into a table of size 1000.

Given Employee ID = 1234, compute the hash. Show all steps and explain why mid-square offers better dispersion than modulo-based hashing.

9. Another company uses the folding-square method for hashing 8-digit employee IDs.

Given Employee ID = 43127896, compute the hash value using this method and explain the benefits of folding with squaring.

## Message Authentication & Communication Integrity

10. Suppose a Message Authentication Code (MAC) changes after transmission. What are the implications? How does this help detect tampering in communication systems?

11. A bank uses RSA and digital signatures to secure online transactions. How do digital signatures help prevent spoofing and ensure secure communication?

## Cryptographic Principles & Key Exchange

12. Describe the role of key exchange protocols in symmetric key cryptography. Why is this a challenge over insecure channels?

13. Compare and contrast symmetric and asymmetric key cryptography in terms of security, speed, and use cases.

14. Bob sends an encrypted message using Alice's public key. What cryptographic principle is used here, and how does this ensure confidentiality?

## Cybersecurity Tools & Performance

15. Your organization is experiencing pop-up ads, poor performance, and unknown installations. Identify the likely malware types and suggest tools/protocols (anti-spyware, system tuning) to mitigate the issue.

16. You are evaluating two antivirus solutions-one offers real-time scanning but slows the system, and the other offers fast scans but no real-time protection. For a business with sensitive data, which is better and why?

17. You are using a system tuning tool that reduces CPU usage by 25%. If the original usage was 80%, what's the new usage? Is this efficient for high-security operations?

18. A phishing detection tool flags 15 out of 100 emails as suspicious. If 12 are actually malicious, calculate its accuracy and suggest one improvement.

## Classical Ciphers

19. Encrypt the message "INFORMATION" using a columnar transposition cipher with the keyword "KEY".
   - Write the message in rows under columns labeled by keyword letters.
   - Rearrange columns alphabetically and read column-wise for the ciphertext.